

Sayan Debnath

Python Backend Engineer — AI Systems — FastAPI

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SUMMARY

Software engineer with hands-on experience building production-grade AI and web systems. Specialized in React and Python (FastAPI), with strong expertise in document intelligence, semantic search, and LLM-powered workflows. Proven ability to design scalable APIs, own features end-to-end, and deliver real-world impact.

PROJECTS

- **DocMind AI** — Built a production-grade AI document intelligence platform using FastAPI, React, and LLMs, supporting section-aware ingestion of large PDF/DOCX files, citation-backed RAG, and agentic multi-step extraction workflows, reducing document search and analysis time by ~60%. [Link](#)
- **Website Search Engine** — Implemented automated content crawling, text chunking, and embedding pipelines, improving search relevance and reducing manual information lookup by ~70%. [Link](#)
- **AI Amazon Listing Generator** — Built an AI-powered Amazon listing generator using Llama 3 served through Ollama to create SEO-optimized titles, bullet points, descriptions, and ad variants, reducing manual listing creation time by ~80%. [GitHub](#)

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript

Frameworks: React.js, Next.js, FastAPI, Flask, Codeigniter

Backend & AI Systems: REST APIs, GraphQL, WebSockets, PostgreSQL, Redis, RAG, Vector Databases

Cloud & DevOps: GCP, Azure, CI/CD, Docker Compose, AWS Lambda

Libraries & Tools: Git, Postman, Pandas, NumPy, OpenCV, Redux, Tailwind CSS

WORK EXPERIENCE

Software Engineer

Smarter Codes, Haryana, India

Aug 2025 - Mar 2026

- Designed and implemented an AI research engine using Python (FastAPI) and vector similarity search for large document corpora.
- Maintained 5+ critical backend microservices using Docker, ensuring ~99.9% uptime and supporting seamless scaling to accommodate increasing user demand.
- Developed 3+ React-based dashboards to monitor AI-driven document processing workflows, enabling real-time tracking of 60+ document processing events per day and improving operational visibility.

Software Developer

Cogzin Technologies, West Bengal, India

Nov 2024 - Aug 2025

- Improved data collection reliability and system integration by optimizing backend processes, reducing data sync errors and improving response processing time by ~20%.
- Designed and integrated 5+ core platform modules including investor profiling, startup listing, recommendation logic, and analytics visualization using modern web technologies.
- Orchestrated continuous integration using Jenkins, enhancing code quality and shortening feedback loops by ~20%, ensuring rapid detection of integration issues, which previously delayed release cycles.

Web Developer (Intern)

Groveus Informatics Pvt.Ltd, West Bengal, India

July 2024 - Dec 2024

- Developed 10+ dynamic web pages using PHP (CodeIgniter) with MySQL database integration, enabling efficient data storage, retrieval, and CRUD operations.
- Implemented clean MVC architecture and database-driven workflows, supporting scalable code structure and maintainability across multiple modules.
- Optimized database queries and application logic, improving page load performance by 25–30% and ensuring smoother user interactions.

EDUCATION

Indira Gandhi National Open University

Bachelor of Computer Applications (Distance Education) — Ongoing

Maidan Garhi, New Delhi, India

2025 - 2028

North Bengal University

Bachelor of Arts (English)

Siliguri, West Bengal, India

2022 - 2025